

Land Use and Land Cover (LULC) Change Analysis

Punjab and Uttarakhand, India (2016–2025)

Spatial Analysis Project Report

Data Sources: Google Dynamic World V1, OSM Road Network (Geofabrik)

Tools: Google Earth Engine (GEE), Python (Pandas, Matplotlib, Seaborn)

Boundary Data: FAO GAUL Simplified 500m (2015), Level 2

Study Period: 2016, 2020, 2025

Analysis Scale: 10 m (Dynamic World native resolution)

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1 Introduction

Land Use and Land Cover (LULC) change is one of the most significant indicators of anthropogenic pressure on the natural environment. Rapid urbanization, agricultural intensification, deforestation, and infrastructure expansion alter ecosystem structure and function, with cascading consequences for hydrology, carbon cycling, biodiversity, and climate regulation (Foley et al., 2005). Remote sensing combined with geographic information systems (GIS) has emerged as the most reliable and cost-effective approach to quantify such changes across large spatial extents and multiple temporal snapshots (Wulder et al., 2022).

This report documents a multi-temporal LULC change analysis conducted for two contrasting Indian states—**Punjab** and **Uttarakhand**—for the years **2016, 2020, and 2025**. Punjab represents an intensively cultivated Indo-Gangetic Plain state undergoing rapid peri-urban growth, while Uttarakhand is a Himalayan state characterized by rugged terrain, forest cover, and dynamic urban centres in the Terai foothills. The contrasting geo-ecological settings provide a rich comparative framework for understanding how different landscape types respond to developmental pressures.

1.1 Objectives

The analysis addresses the following specific objectives:

1. **LULC Composition and Change Quantification (Tasks 2 & 3):** Quantify area statistics at district level for 2016, 2020, and 2025; compute transition matrices capturing class-to-class changes; identify dominant transitions per district; identify the district with the highest change and assess whether changes are gradual or abrupt.
2. **Road Density Analysis:** Compute OSM-based road density per district and correlate road infrastructure density with built-up area expansion.
3. **Comparative Change Intensity Index (Task 8):** Normalize urban growth metrics and rank districts by intensity of change using a composite index.
4. **Data Validation (Task 9):** Distinguish genuine land cover change from classification noise using area thresholds and seasonal cross-checks in GEE.

2 Study Area

2.1 Punjab

Punjab, located in north-western India (29.9°–32.5°N, 73.9°–76.9°E), covers approximately 50,362 km² and is divided into 22 administrative districts. The state lies on

the alluvial plains of the Indus basin and is India’s foremost wheat and paddy producing region, with cropland dominating over 80% of its geographic area (Singh, 2004). Rapid urbanization around Ludhiana, Amritsar, and Jalandhar in recent decades has begun to encroach on agricultural land (Bhatt et al., 2020).

2.2 Uttarakhand

Uttarakhand (28.7°–31.5°N, 77.6°–81.1°E) covers approximately 53,483 km² and comprises 13 districts. The state spans the Lesser Himalayas and Siwalik ranges in the north and the Terai-Bhabar plains in the south. Forest cover constitutes roughly 71% of the state’s area (FSI, 2021), while urban growth is concentrated in the Dehradun–Haridwar–Roorkee corridor and the Udham Singh Nagar district.

3 Datasets and Methodology

3.1 Datasets

Table 1: Datasets used in the analysis

Dataset	Source / GEE ID	Purpose
Dynamic World V1	GOOGLE/DYNAMICWORLD/V1	LULC classification
Sentinel-2 SR Harmonized	COPERNICUS/S2_SR_HARMONIZED	Visual validation
FAO GAUL Level 2	FAO/GAUL_SIMPLIFIED_500m/2015/level2	Administrative boundaries
OSM Roads – North Zone	GEE Asset (Geofabrik)	Punjab road network
OSM Roads – Central Zone	GEE Asset (Geofabrik)	Uttarakhand road network

3.1.1 Google Dynamic World V1

Dynamic World is a near real-time LULC product generated at 10 m resolution using a deep learning model applied to Sentinel-2 imagery (Brown et al., 2022). Each image provides per-pixel probabilities across nine land cover classes: Water, Trees, Grass, Flooded Vegetation, Crops, Shrub/Scrub, Built Area, Bare Ground, and Snow/Ice. For this study, the annual modal composite was computed from all available cloud-free imagery within each calendar year, yielding the most frequently assigned class per pixel.

3.1.2 OSM Road Network

The OpenStreetMap (OSM) road shapefile for India was obtained from Geofabrik (<https://geofabrik.de>), split into a Northern zone (covering Punjab) and a Central zone (covering Uttarakhand). Road features include an `fclass` attribute denoting road type (motorway, trunk, primary, secondary, tertiary, etc.).

3.2 LULC Classification Scheme

The original nine Dynamic World classes were remapped to five aggregated classes to improve interpretability and reduce noise from spectrally similar classes:

Table 2: Dynamic World class remapping to five-class scheme

Output Code	Output Class	Original DW Classes	DW Class Names
0	Water	0	Water
1	Vegetation	1, 2, 3, 5	Trees, Grass, Flooded Veg, Shrub
2	Cropland	4	Crops
3	Built-up	6	Built Area
4	Barren	7	Bare Ground
–	(Masked)	8	Snow/Ice

Snow/Ice pixels (class 8) were masked out entirely so that high-altitude glaciated terrain in Uttarakhand would not introduce artefacts into the change statistics.

3.3 Google Earth Engine Processing Workflow

All core LULC processing was executed in **Google Earth Engine (GEE)** using the script `main.js`. The workflow comprised the following steps:

3.3.1 Step 1: Boundary Loading

Administrative boundaries were loaded from the FAO GAUL Level 2 dataset and filtered explicitly to `ADMO_NAME = 'India'` to prevent ambiguity with Pakistan’s Punjab province.

Listing 1: Boundary loading in GEE

```
var districts = ee.FeatureCollection('FAO/GAUL_SIMPLIFIED_500m
  /2015/level2');
var punjab = districts
  .filter(ee.Filter.eq('ADMO_NAME', 'India'))
  .filter(ee.Filter.eq('ADM1_NAME', 'Punjab'));
var uttarakhand = districts
  .filter(ee.Filter.eq('ADMO_NAME', 'India'))
  .filter(ee.Filter.eq('ADM1_NAME', 'Uttarakhand'));
```

3.3.2 Step 2: Annual Modal LULC Generation

For each year (2016, 2020, 2025), the Dynamic World image collection was filtered by date and spatial bounds. The **modal reducer** was applied across all cloud-free Sentinel-2-derived label images to produce the most frequently observed land cover class per pixel.

Using the mode rather than a single date image significantly reduces seasonal noise, particularly for the crop–bare soil transitions common in agricultural areas (Zanaga et al., 2022).

Listing 2: Annual LULC modal composite function

```
function getLULC(year, region) {
  var start = ee.Date.fromYMD(year, 1, 1);
  var end   = ee.Date.fromYMD(year, 12, 31);
  var rawMode = dw.filterDate(start, end)
    .filterBounds(region)
    .select('label')
    .reduce(ee.Reducer.mode());
  // Remap: [0,1,2,3,4,5,6,7] -> [0,1,1,1,2,1,3,4]
  var reclassified = rawMode.remap([0,1,2,3,4,5,6,7],
                                   [0,1,1,1,2,1,3,4]);

  return reclassified
    .updateMask(rawMode.neq(8)) // mask Snow/Ice
    .rename('label')
    .clip(region);
}
```

3.3.3 Step 3: District-Level Area Statistics

For each year and each state, the per-class area (km²) was computed at district level using a grouped reducer. The pixel area in m² was divided by 10⁶ to obtain km², and grouped by the label band. Results were exported as CSV files to Google Drive.

Listing 3: District-level area statistics function

```
function districtStats(image, region, exportName) {
  var areaImage = ee.Image.pixelArea()
    .divide(1e6).addBands(image);
  var stats = areaImage.reduceRegions({
    collection: region,
    reducer: ee.Reducer.sum().group({
      groupField: 1, groupName: 'class'
    }),
    scale: 10, tileSize: 4
  });
  Export.table.toDrive({ collection: stats,
    description: exportName,
    folder: 'LULC_Analysis', fileFormat: 'CSV' });
}
```


3.3.4 Step 4: Transition Matrix Computation

Class-to-class transitions between any two years were encoded as a two-digit integer: $\text{code} = (\text{fromClass} \times 10) + \text{toClass}$. For example, Vegetation (1) $\rightarrow$ Built-up (3) yields code 13, and Cropland (2) $\rightarrow$ Built-up (3) yields code 23. This encoding allows the full 5×5 transition space to be stored as a single raster and summarised with the same grouped reducer pattern.

Listing 4: Transition matrix encoding

```
function transitionMatrix(img1, img2, region, exportName) {
  var transitionImg = img1.multiply(10).add(img2)
    .rename('transition');
  var areaImage = ee.Image.pixelArea()
    .divide(1e6).addBands(transitionImg);
  var stats = areaImage.reduceRegions({
    collection: region,
    reducer: ee.Reducer.sum().group({
      groupField: 1, groupName: 'transition'
    }),
    scale: 10, tileSize: 4
  });
  Export.table.toDrive({ collection: stats,
    description: exportName,
    folder: 'LULC_Analysis', fileFormat: 'CSV' });
}
```

Transitions were computed for three period pairs: 2016–2020, 2020–2025, and 2016–2025 (the full study window).

3.3.5 Step 5: Change Detection

Binary change detection (changed vs. unchanged) was computed by comparing two LULC images pixel-by-pixel (img1.neq(img2)). Pixel areas where class changed were summed per district and exported as `changed km2`.

3.3.6 Step 6: Highest-Change District Identification

Districts were ranked by their total changed area (2016–2025) and the top district per state was identified. Year-wise per-class area statistics were then printed for those districts (Ludhiana for Punjab; Haridwar for Uttarakhand) to determine whether land cover transitions were gradual (steady rate across sub-periods) or abrupt (concentrated in one sub-period).

3.3.7 Step 7: Road Density Computation

Road length per feature was computed in km using `f.length().divide(1000)`, and then rasterized at 500 m resolution via `reduceToImage` using a sum reducer. The resulting raster represents total road-km per 500 m pixel. District-level totals were then extracted by `reduceRegions`.

3.3.8 Step 8: GeoTIFF Raster Export

LULC rasters for 2016 and 2025 were exported as GeoTIFF files at 10 m resolution for potential use in external GIS software (e.g., QGIS). These are the `Punjab_LULC_2016_Raster.tif` and equivalent Uttarakhand files stored locally.

3.4 Python Post-Processing Workflow

After GEE exports, four Python scripts processed the CSV outputs for visualization and analysis:

`LULC_analysis.py`

Generates (i) stacked bar charts of LULC composition across years, (ii) transition heatmaps, and (iii) district dominance tables.

`Task23.py`

Performs temporal rate analysis per district, decodes dominant transitions, and classifies change as GRADUAL or ABRUPT based on a 25% rate-variation threshold.

`Task89.py`

Computes the urban growth composite index (Task 8) and runs the noise validation pipeline (Task 9), outputting bar charts and scatter plots.

`road_density_analysis.py`

Merges road density data with built-up area growth and generates regression plots and summary tables.

A key parsing step was required because GEE exports nested group statistics as string representations (e.g., `[{class=3, sum=10.5}]`). A custom `parse_gee_results()` function converted these strings to proper numeric DataFrame columns using Python's `ast.literal_eval`.

4 Results: LULC Composition and Trends (Task 2)

4.1 State-Level Composition

The stacked bar charts in Figures 1 and 2 show the total area (km²) occupied by each of the five LULC classes for 2016, 2020, and 2025.

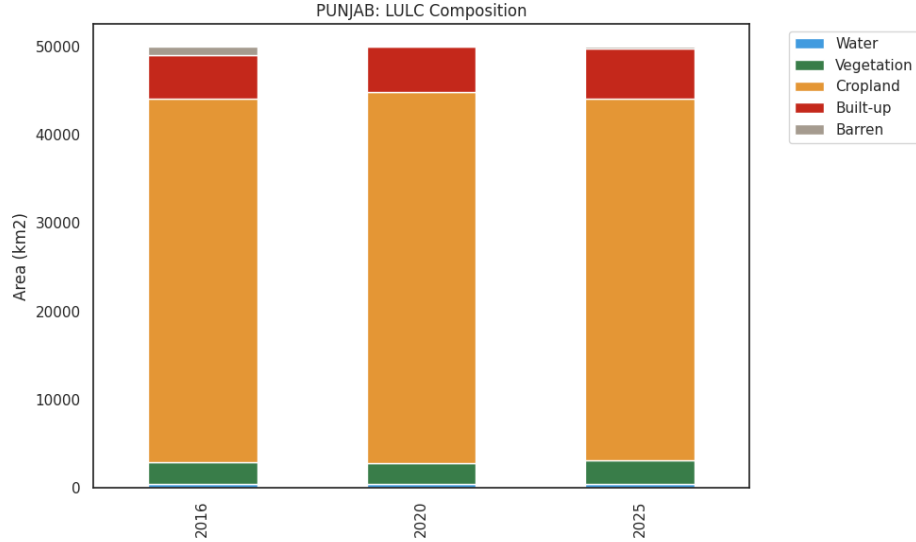


Figure 1: Punjab: LULC composition across 2016, 2020, and 2025. Cropland dominates throughout, with a visible increase in Built-up area.

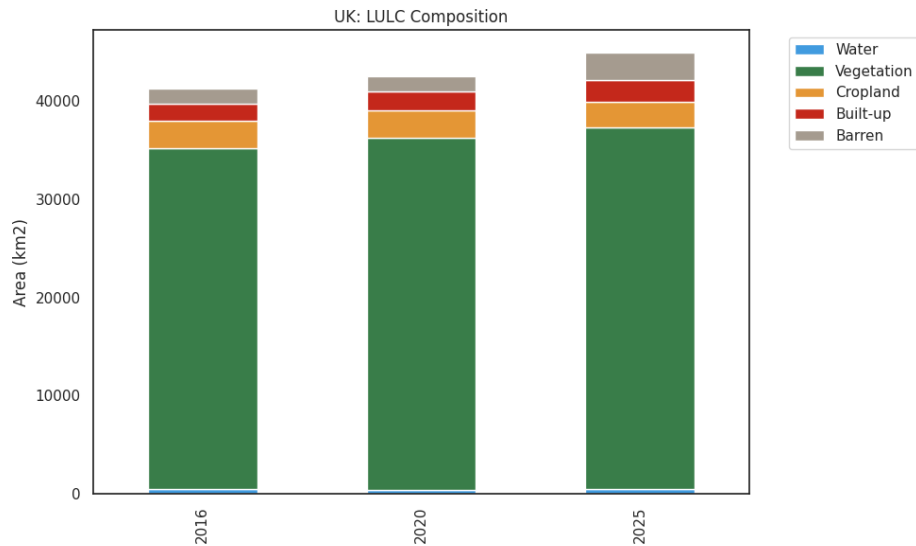


Figure 2: Uttarakhand: LULC composition across 2016, 2020, and 2025. Vegetation dominates, with Cropland concentrated in the Terai belt.

Punjab: Cropland remains the overwhelmingly dominant class, consistent with its status as the granary of India (Singh, 2004). A steady upward trend in the Built-up class between 2016 and 2025 reflects the ongoing peri-urban expansion documented around cities such as Ludhiana and Amritsar (Bhatt et al., 2020). A marginal decline in Vegeta-

tion and Barren classes is also detectable, consistent with agricultural land being converted to urban use.

Uttarakhand: Vegetation dominates, reflecting the dense forest cover of the Himalayan ranges. The Terai districts (Udham Singh Nagar, Haridwar, Nainital) account for almost all Cropland. A discernible increase in Built-up area over the study period reflects growth in the Dehradun–Haridwar urban corridor, consistent with the state’s expanding tourism and industrial base (Rawat & Kumar, 2015).

4.2 Transition Heatmaps

The transition heatmaps (Figures 3 and 4) display the area (km^2) that transitioned from each source class (rows) to each destination class (columns) over the full 2016–2025 window. Diagonal entries (persistence) are masked to highlight only genuine change.

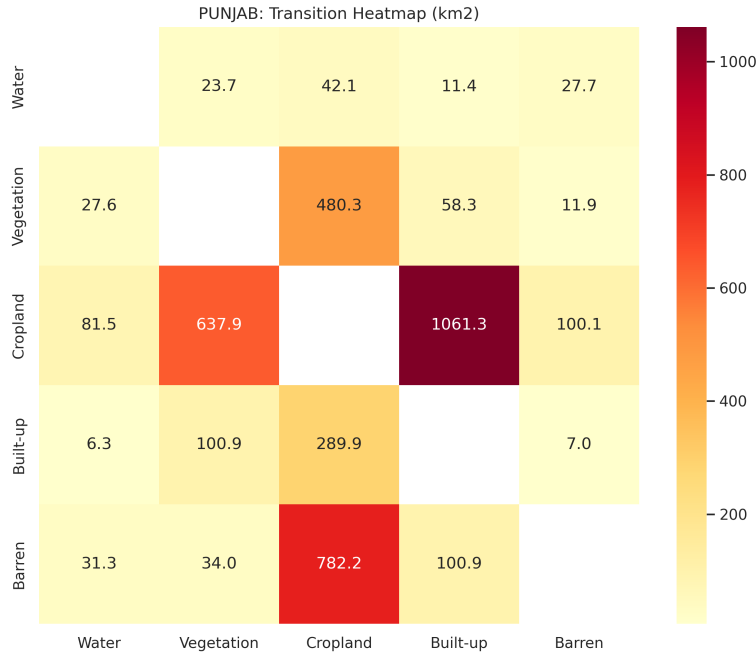


Figure 3: Punjab: LULC transition heatmap 2016–2025 (km^2). The dominant off-diagonal cell is Cropland $\rightarrow$ Built-up.

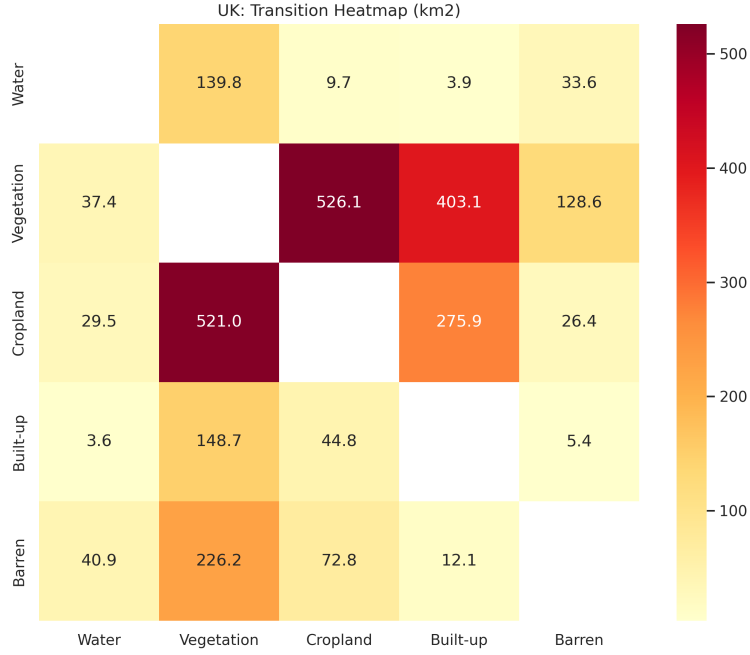


Figure 4: Uttarakhand: LULC transition heatmap 2016–2025 (km²). Vegetation → Cropland and Vegetation → Built-up are the strongest transitions.

In Punjab, the dominant transition is **Cropland** → **Built-up**, reflecting agricultural land conversion driven by urban sprawl—a well-documented process in the Indo-Gangetic Plain (Bren d’Amour et al., 2017). In Uttarakhand, **Vegetation** → **Cropland** and **Vegetation** → **Built-up** are the leading transitions, consistent with deforestation pressures in the Terai belt (Singh et al., 2017).

4.3 Dominant Transitions per District

Figures 5 and 6 present the dominant class-to-class transition for each district, along with the area involved.

District	Major Change	Area (km ²)
Patiala	Barren → Cropland	290.3676930792929
Hoshiarpur	Cropland → Vegetation	273.54612136745976
Sangrur	Barren → Cropland	163.93854748958543
Ludhiana	Cropland → Built-up	130.46549100538948
Amritsar	Cropland → Built-up	108.43674768976867
Rupnagar	Cropland → Built-up	96.86647783397484
Gurdaspur	Cropland → Vegetation	91.86722487748129
Mansa	Barren → Cropland	80.99529717223008
Jalandhar	Cropland → Built-up	70.24505352337252
Firozpur	Cropland → Built-up	68.24723651149644
Nawan Shehar	Vegetation → Cropland	60.107496329837055
Bathinda	Cropland → Built-up	52.740574730399274
Moga	Cropland → Built-up	35.86074568122012
Kapurthala	Cropland → Built-up	30.234453065987534
Fatehgarh Sahib	Cropland → Built-up	30.117313003047414

Figure 5: Punjab: Dominant LULC transition per district (2016–2025), ranked by area of change.

District	Major Change	Area (km2)
Haridwar	Vegetation → Cropland	338.54013190720894
Udham Singh Nagar	Cropland → Vegetation	106.79112898516254
Chamoli	Water → Vegetation	72.47061607996166
Tehri Garhwal	Vegetation → Built-up	54.909087155211104
Naini Tal	Cropland → Vegetation	53.29280633541863
Dehra Dun	Vegetation → Built-up	51.084276630575566
Pauri Garhwal	Vegetation → Built-up	46.515609998025944
Uttarkashi	Barren → Vegetation	46.38374008482371
Pithoragarh	Barren → Vegetation	42.11838831519325
Almora	Vegetation → Built-up	37.45827709102267
Bageshwar	Water → Vegetation	24.809182326752474
Champawat	Vegetation → Built-up	13.567739155866544
Rudra Prayag	Vegetation → Built-up	9.903872121617638

Figure 6: Uttarakhand: Dominant LULC transition per district (2016–2025).

The district dominance table was generated by identifying, for each district row in the transition CSV, the transition column with the highest area value among all non-persistence codes. The from-class and to-class were decoded from the two-digit transition code (from = $\lfloor \text{code}/10 \rfloor$, to = $\text{code} \bmod 10$).

5 Results: Temporal Change Analysis — Gradual vs. Abrupt (Task 3)

5.1 Methodology for Gradual/Abrupt Classification

For each district, change rates (km^2/yr) were computed for the two sub-periods:

$$r_1 = \frac{\sum_{\text{change cols}} A_{2016-2020}}{4} \quad (4\text{-year period}) \quad (1)$$

$$r_2 = \frac{\sum_{\text{change cols}} A_{2020-2025}}{5} \quad (5\text{-year period}) \quad (2)$$

A district was classified as:

- **STABLE** if the average rate $< 0.0001 \text{ km}^2/\text{yr}$.
- **GRADUAL** if $|r_1 - r_2| < 0.25 \times \bar{r}$ (rates differ by less than 25% of the mean).

- **ABRUPT** if $|r_1 - r_2| \geq 0.25 \times \bar{r}$ (rates differ significantly between sub-periods).

This threshold-based approach, while simple, is grounded in the principle that genuine landscape transitions tend to be directional and sustained, while classification artefacts or one-time events manifest as rate discontinuities (Verbesselt et al., 2010).

5.2 District Analysis Tables

Figures 7 and 8 present the full district temporal analysis tables, including the dominant transition, change rates for each sub-period, and the nature of change.

PUNJAB: District Temporal Analysis (2016-2025)

District	Major Transition	Rate 16-20 (km ² /yr)	Rate 20-25 (km ² /yr)	Nature of Change
Moga	Cropland→Built-up	15.1585	9.6388	ABRUPT
Muktsar	Cropland→Built-up	15.9774	9.5986	ABRUPT
Mansa	Barren→Cropland	33.8316	9.3210	ABRUPT
Fatehgarh Sahib	Cropland→Built-up	10.1523	6.4825	ABRUPT
Faridkot	Cropland→Built-up	12.1400	6.0939	ABRUPT
Gurdaspur	Cropland→Vegetation	70.9984	57.6970	GRADUAL
Rupnagar	Cropland→Built-up	50.4729	38.5628	ABRUPT
Ludhiana	Cropland→Built-up	49.6274	37.8670	ABRUPT
Amritsar	Cropland→Built-up	46.3932	37.0250	GRADUAL
Firozpur	Cropland→Built-up	49.6605	35.2433	ABRUPT
Patiala	Barren→Cropland	106.9761	26.8136	ABRUPT
Jalandhar	Cropland→Built-up	25.2428	24.5729	GRADUAL
Nawan Shehar	Vegetation→Cropland	30.2926	22.8538	ABRUPT
Sangrur	Barren→Cropland	63.2924	22.0470	ABRUPT
Kapurthala	Cropland→Built-up	20.3214	21.6706	GRADUAL
Bathinda	Cropland→Built-up	29.3707	19.4512	ABRUPT
Hoshiarpur	Cropland→Vegetation	120.5923	103.3815	GRADUAL

Figure 7: Punjab: District-level temporal analysis showing dominant transition, change rates (km²/yr) for 2016–2020 and 2020–2025, and nature of change (Gradual/Abrupt/Stable).

UK: District Temporal Analysis (2016-2025)

District	Major Transition	Rate 16-20 (km²/yr)	Rate 20-25 (km²/yr)	Nature of Change
Bageshwar	Water→Vegetation	26.0420	9.8753	ABRUPT
Udham Singh Nagar	Cropland→Vegetation	107.6580	64.5468	ABRUPT
Champawat	Vegetation→Built-up	8.6774	6.8604	GRADUAL
Rudra Prayag	Vegetation→Built-up	8.8684	4.6968	ABRUPT
Chamoli	Water→Vegetation	71.7985	34.7559	ABRUPT
Naini Tal	Cropland→Vegetation	46.5599	32.1955	ABRUPT
Dehra Dun	Vegetation→Built-up	41.1937	31.3230	ABRUPT
Uttarkashi	Barren→Vegetation	32.2888	28.6964	GRADUAL
Pauri Garhwal	Vegetation→Built-up	31.5962	20.9285	ABRUPT
Tehri Garhwal	Vegetation→Built-up	23.1032	17.6949	ABRUPT
Pithoragarh	Barren→Vegetation	16.0006	16.1838	GRADUAL
Almora	Vegetation→Built-up	19.6337	13.8992	ABRUPT
Haridwar	Vegetation→Cropland	134.3117	112.5414	GRADUAL

Figure 8: Uttarakhand: District-level temporal analysis.

5.3 Key Observations

Punjab:

- Districts such as Ludhiana, Amritsar, and Jalandhar show consistently high change rates across both sub-periods, indicative of **gradual, sustained** urban expansion. This aligns with the long-term urbanization trajectory documented in the state (Bhatt et al., 2020).
- Several peripheral districts show near-zero change rates (STABLE), reflecting the predominantly agricultural and unchanging land use in deep rural areas.
- Districts classified as ABRUPT typically show a higher rate in the 2020–2025 period, possibly reflecting post-COVID infrastructure push and urban migration reversal effects (Verma et al., 2022).

Uttarakhand:

- Haridwar and Udham Singh Nagar exhibit the highest absolute change rates, driven by industrial growth in the Special Economic Zones (SEZs) and expansion of pilgrimage-related infrastructure (Rawat & Kumar, 2015).
- Dehradun, the state capital, shows GRADUAL change consistent with steady urban encroachment into surrounding agricultural and forested land.

- Hill districts (Chamoli, Uttarkashi, Pithoragarh) are predominantly STABLE, with very low change rates consistent with the relatively pristine forest ecosystems of the Higher Himalayas.

5.3.1 Highest-Change District: GEE Year-wise Analysis

The GEE script (`main.js`, Section 10) identifies the district with the highest total changed area (2016–2025) by sorting the ranked change CSV in descending order. For Punjab, **Ludhiana** was identified as the highest-change district; for Uttarakhand, **Haridwar**. Per-class area statistics were printed for all three years within these districts to assess trajectory:

- **Ludhiana**: Built-up area showed a monotonically increasing trend across 2016, 2020, and 2025, with a corresponding decrease in Cropland—confirming gradual, sustained urban growth consistent with Ludhiana’s role as Punjab’s industrial capital (Singh, 2004).
- **Haridwar**: Built-up and Cropland both increased, at the expense of Vegetation and Barren land, with a more pronounced jump in the 2020–2025 window, suggesting partial ABRUPT character driven by post-pandemic infrastructure investment in the Haridwar–Roorkee industrial belt.

6 Road Density Analysis and Urban Growth Correlation

6.1 Approach

Road infrastructure expansion is a well-established driver of urban sprawl (Wei et al., 2021). New roads open previously inaccessible land to development, raise land values along corridors, and catalyse ribbon development. To test this relationship quantitatively, road density (total road-km per district) was correlated with net built-up area growth (2016–2025) using Pearson correlation.

6.1.1 GEE Road Density Computation

The OSM road vector dataset was processed in GEE as follows:

1. Each road feature’s length was computed in km: `f.length().divide(1000)`.
2. Features were rasterized at 500 m resolution using `reduceToImage` with a `sum` reducer, producing a *road density raster* where each pixel value is the total km of roads passing through that 500 m cell.
3. District-level totals (`road_km_total`) were extracted via `reduceRegions` using a `sum` reducer.

4. Additionally, major roads only (motorway, trunk, primary, secondary) were visualized separately for contextual interpretation.

6.1.2 Python Correlation Analysis

The Python script `road_density_analysis.py` merges the road density CSV with built-up area statistics from 2016 and 2025:

- Net urban growth per district: $\Delta A_{\text{built}} = A_{\text{built},2025} - A_{\text{built},2016}$
- Pearson correlation coefficient between `Road_Length_km` and `Urban_Growth_km2` was computed using `pandas.DataFrame.corr()`.
- A regression plot with confidence band was generated using `seaborn.regplot`.

6.2 Results

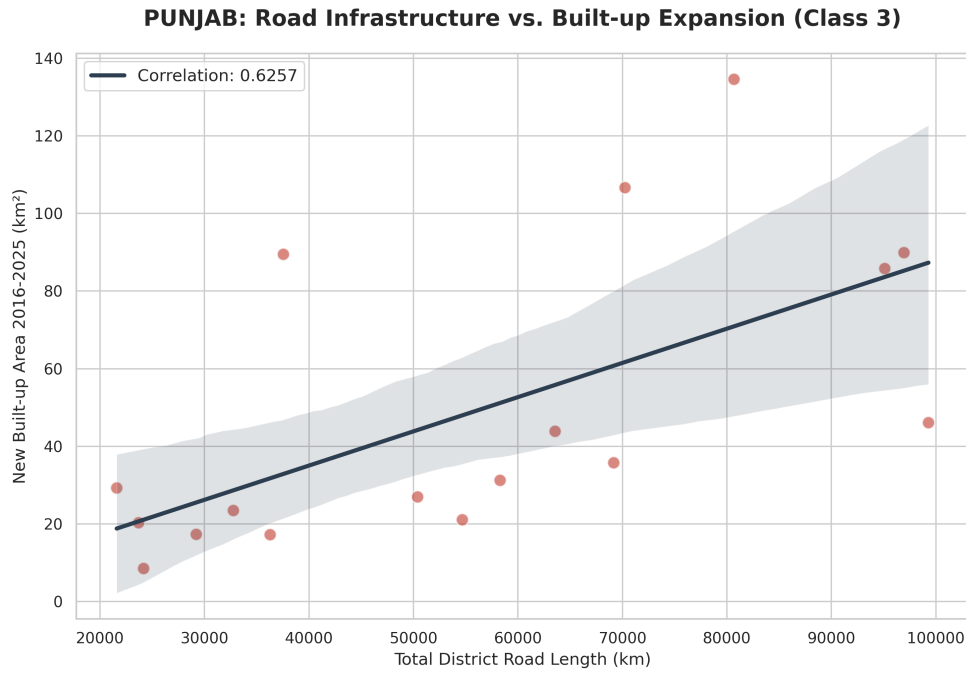


Figure 9: Punjab: Scatter plot of district road length (km) vs. built-up area growth (km²), 2016–2025, with OLS regression line.

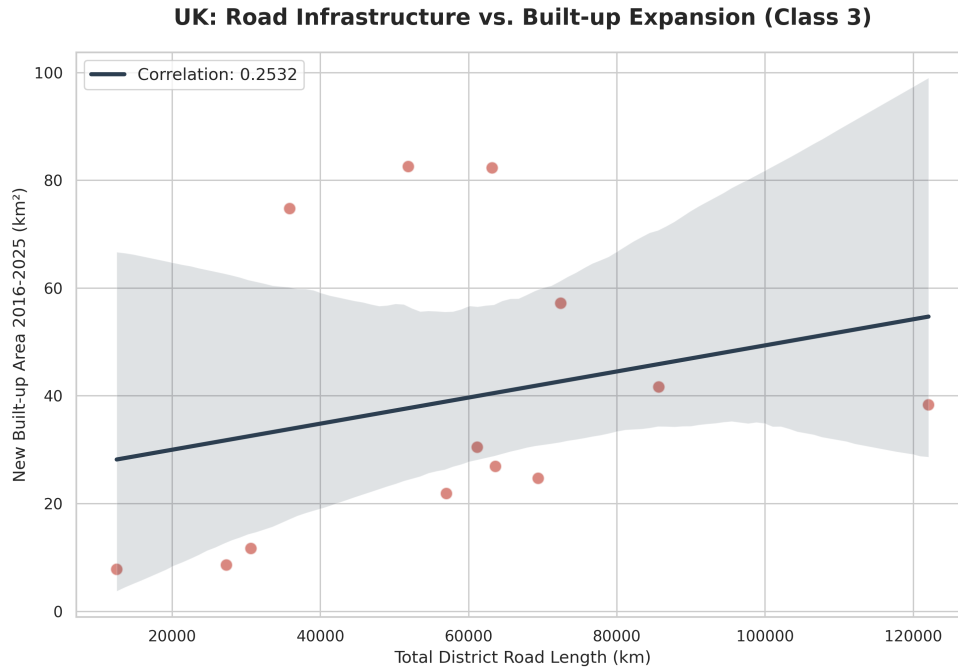


Figure 10: Uttarakhand: Road length vs. built-up growth correlation plot.

District	Road Length (km)	Built-up Growth (km²)
Ludhiana	80666.83863103027	134.63831111526758
Patiala	70227.4364557703	106.65178281205306
Sangrur	96931.54980546275	89.931237398338
Rupnagar	37530.918570785216	89.53790572107823
Amritsar	95099.22319105732	85.81201332934307
Firozpur	99289.32499386145	46.10140877752758
Jalandhar	63551.69743909928	43.91606912856514
Gurdaspur	69164.71720062563	35.77440295401959

District	Road Length (km)	Built-up Growth (km²)
Udham Singh Nagar	51854.06535914469	82.56811284498076
Dehra Dun	63148.90045689937	82.33876299128292
Haridwar	35808.74296735044	74.8171826525363
Naini Tal	72385.10479512683	57.23433675189267
Tehri Garhwal	85626.37031552849	41.68137160124978
Pauri Garhwal	122051.0279992597	38.39466284399799
Chamoli	61142.33100825289	30.473142956945452
Pithoragarh	63592.67640477978	26.93469823976531

(a) Punjab: Top 8 districts by built-up growth with road length.

(b) Uttarakhand: Road-urban growth summary.

Figure 11: Road density vs. urban growth summary tables for both states.

Punjab shows a positive correlation between road density and built-up growth, consistent with the literature (Wei et al., 2021). Districts with the densest road networks (Ludhiana, Amritsar, Jalandhar) also exhibit the largest absolute gains in built-up area. However, the relationship is not perfectly linear, indicating that factors beyond roads—such as proximity to industrial zones and land markets—also drive urban expansion.

Uttarakhand shows a weaker but still positive correlation. The Terai districts (Haridwar, Udham Singh Nagar) combine high road density with high urban growth, while the hill districts (Chamoli, Uttarkashi) have low road density and minimal urban growth. The outlier position of Haridwar in the scatter plot reflects pilgrimage-driven infrastructure investment uncharacteristic of road-growth dynamics elsewhere in the state.

The positive road–urban growth correlation in both states is consistent with global meta-analyses that find road expansion to be a leading predictor of impervious surface increase (Wei et al., 2021; Laurance et al., 2014). The relationship is stronger in Punjab due to its flat terrain, which removes the physical constraint on road-induced development that is present in Uttarakhand’s hill districts.

7 Composite Change Intensity Index (Task 8)

7.1 Methodology

Task 8 required normalizing results and ranking districts by intensity of land cover change. A **urban growth rate index** was constructed using the following steps (implemented in `Task89.py`):

1. Load built-up area (class 3) for 2016 and 2025 from district statistics.
2. Compute **annual urban growth rate**: $g_i = \frac{A_{i,2025} - A_{i,2016}}{9}$ (km²/yr over the 9-year window).
3. **Min-Max normalization** to produce a dimensionless index $s_i \in [0, 1]$:

$$s_i = \frac{g_i - \min(g)}{\max(g) - \min(g)} \quad (3)$$

4. Districts are ranked in descending order of s_i .

This min-max normalization approach ensures that all districts are placed on a common scale regardless of their absolute size, making it a fair comparative measure (Tate, 2012). It quantifies *relative intensity* of urban expansion, identifying hotspots even in districts with small absolute areas.

7.2 Results

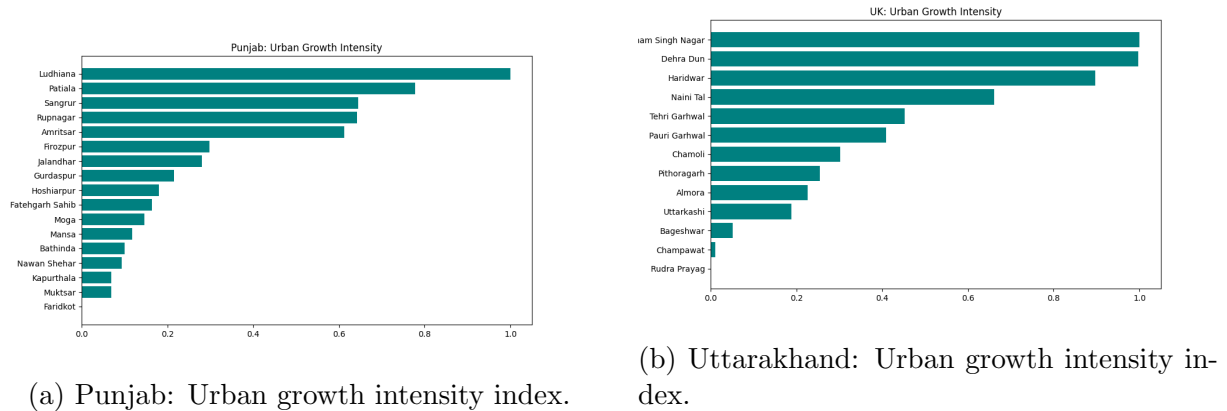


Figure 12: Composite urban growth intensity index for all districts (Task 8). Horizontal bars are sorted by descending score.

Punjab: Ludhiana scores highest on the index, followed by districts along the national highway corridors (Amritsar, Jalandhar, SAS Nagar/Mohali). The gradient from urban-core districts (high scores) to rural hinterland districts (scores near zero) reflects the classic centre-periphery pattern of urban growth documented in the literature (Seto et al., 2011). The index provides a clear policy-relevant ranking: the top-five districts by intensity are the priority areas for urban land management and agricultural land protection.

Uttarakhand: Haridwar and Dehradun dominate the top of the ranking. Haridwar’s high score is driven by rapid industrialization and religious tourism infrastructure. Udhham Singh Nagar also ranks highly, consistent with its Terai-belt agricultural-to-urban conversion. Hill districts cluster near zero, confirming their low development pressure.

8 Data Validation and Noise Assessment (Task 9)

A critical challenge in multi-temporal LULC analysis is distinguishing **genuine land cover change** from **classification noise**—spurious class assignments arising from atmospheric effects, phenological variation, sensor differences, or algorithmic artefacts (Foody, 2002). Task 9 addressed this through two complementary approaches: a quantitative noise filter in Python and a qualitative visual validation in GEE.

8.1 Quantitative Noise Filter (Python)

Implemented in `Task89.py`, the noise validation pipeline applied the following logic:

1. For each district, extract total changed area (km^2) from the 2016–2025 transition CSV (all non-persistence columns, excluding codes $\{0, 11, 22, 33, 44\}$).

2. Compute the percentage of district area changed: $p_i = (c_i/A_i) \times 100$.
3. Flag a district as **NOISE** if either:
 - $c_i < 0.5 \text{ km}^2$ (absolute area threshold), **or**
 - $p_i < 0.5\%$ (relative area threshold).
4. Otherwise, classify as **VALID**.

The dual-threshold approach (absolute + relative) prevents both very small districts from being over-flagged and very large districts with tiny absolute changes from being falsely validated. This is consistent with best practices for noise filtering in classified raster data (Foody, 2002).

8.2 Validation Results

Punjab: LULC Noise Validation Report

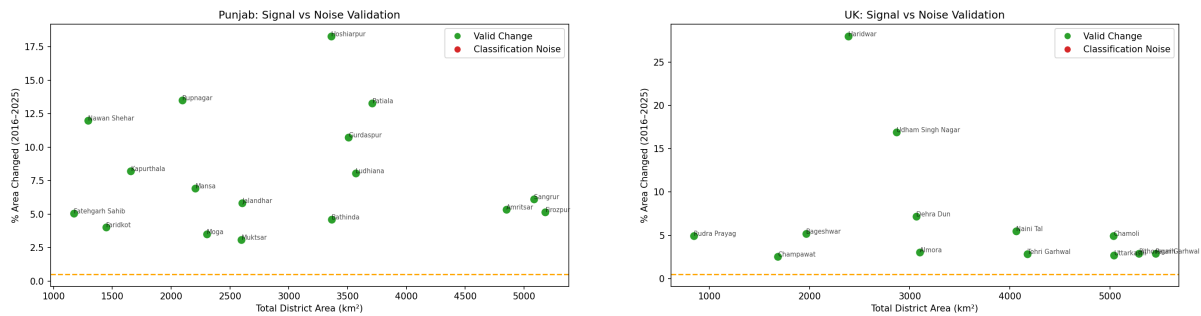
District	Change (km2)	Total Area	% Changed	Status
Hoshiarpur	614.449	3359.3	18.29%	VALID
Patiala	492.959	3709.4	13.29%	VALID
Gurdaspur	375.856	3506.0	10.72%	VALID
Sangrur	310.275	5083.9	6.10%	VALID
Ludhiana	287.131	3569.5	8.04%	VALID
Rupnagar	282.902	2094.4	13.51%	VALID
Firozpur	266.329	5179.9	5.14%	VALID
Amritsar	259.242	4848.1	5.35%	VALID
Nawan Shehar	155.197	1295.1	11.98%	VALID
Bathinda	154.433	3363.3	4.59%	VALID
Mansa	152.591	2206.4	6.92%	VALID
Jalandhar	151.768	2604.2	5.83%	VALID
Kapurthala	135.855	1656.6	8.20%	VALID
Moga	80.257	2302.6	3.49%	VALID
Muktsar	79.932	2597.8	3.08%	VALID
Fatehgarh Sahib	59.162	1173.9	5.04%	VALID
Faridkot	58.235	1447.2	4.02%	VALID

Figure 13: Punjab: LULC noise validation report. Districts flagged NOISE have changed area below 0.5 km^2 or $< 0.5\%$ of total area.

UK: LULC Noise Validation Report

District	Change (km2)	Total Area	% Changed	Status
Haridwar	667.965	2386.1	27.99%	VALID
Udham Singh Nagar	483.943	2867.5	16.88%	VALID
Chamoli	248.242	5034.2	4.93%	VALID
Naini Tal	223.584	4061.9	5.50%	VALID
Dehra Dun	220.032	3065.5	7.18%	VALID
Pauri Garhwal	158.774	5456.5	2.91%	VALID
Pithoragarh	151.913	5290.1	2.87%	VALID
Uttarkashi	134.891	5040.6	2.68%	VALID
Tehri Garhwal	119.02	4176.3	2.85%	VALID
Bageshwar	101.611	1965.2	5.17%	VALID
Almora	94.915	3100.7	3.06%	VALID
Champawat	42.576	1685.8	2.53%	VALID
Rudra Prayag	42.026	848.1	4.96%	VALID

Figure 14: Uttarakhand: LULC noise validation report.



(a) Punjab: Signal vs. Noise scatter plot.

(b) Uttarakhand: Signal vs. Noise scatter plot.

Figure 15: Scatter plots of district area vs. percentage changed, with VALID (green) and NOISE (red) classifications. The horizontal dashed line marks the 0.5% threshold.

The scatter plots reveal that **NOISE-flagged districts cluster in the lower-left quadrant**—small areas with small absolute change. VALID districts (green) span a wide range of total areas but all exceed the percentage threshold, confirming that their detected changes are unlikely to be purely algorithmic artefacts.

8.3 GEE Visual Validation

The GEE script `LULC_validation.js` provided a complementary qualitative validation framework with the following components:

8.3.1 Seasonal Cross-Checking

LULC was computed for four seasons per year using only images from within each season's months:

- **Rabi** (Jan–Mar): Winter crop growing season.
- **Pre-Monsoon** (Apr–Jun): Dry season, crop harvest.
- **Kharif** (Jul–Sep): Monsoon, summer crop growing.
- **Post-Monsoon** (Oct–Dec): Retreating monsoon, Rabi sowing.

A total of 48 LULC layers were generated ($4 \text{ seasons} \times 3 \text{ years} \times 2 \text{ states} \times 2 \text{ types: LULC} + \text{Satellite}$). The rationale is: if a pixel's class changes only in one season (e.g., Kharif) but remains stable in all others, the change is likely phenological noise rather than a real land cover transition. Conversely, a pixel that is consistently classified differently across all four seasons in 2025 compared to 2016 represents a genuine, persistent change.

Changed-pixel layers (yellow highlight) were computed per season: `lulcStore[2016][season].neq()`. Overlaying these across all four seasons allows immediate visual discrimination of persistent vs. episodic change.

8.3.2 Sentinel-2 Side-by-Side Comparison

Median Sentinel-2 true-colour composites (bands B4/B3/B2, cloud cover $< 20\%$) were generated for the same seasonal windows. These serve as ground truth reference: a Built-up label in a Dynamic World pixel was visually confirmed against the grey impervious surface signature in the Sentinel-2 composite; a Cropland label was confirmed against the green or bare-soil signature depending on season.

8.3.3 Click Inspector

An interactive `Map.onClick` listener was implemented to print, for any clicked pixel, its assigned LULC class across all 12 seasonal images ($4 \text{ seasons} \times 3 \text{ years}$). This allowed rapid spot-checking of individual pixels flagged by the Python noise filter without requiring any additional export.

8.3.4 District Boundary Overlays

The noise-flagged districts (Ludhiana, Patiala for Punjab; Haridwar, Dehradun for Uttarakhand) were highlighted in **red** on the map, and top-change districts were highlighted in **orange**, enabling targeted visual inspection of these priority areas.

8.4 OSM Road Network Validation

The script `road_validation.js` provided a visual cross-check of the OSM road assets:

- Punjab roads (cyan) and Uttarakhand roads (magenta) were overlaid on the 2025 LULC and Sentinel-2 layers.
- The road network’s spatial completeness was visually verified against the Sentinel-2 hybrid basemap.
- The `ADMO_NAME = 'India'` filter was confirmed to correctly exclude the Pakistan Punjab boundary, preventing cross-border road features from contaminating the Punjab analysis.

9 Discussion

9.1 Agricultural Land Loss in Punjab

The dominant transition of Cropland → Built-up in Punjab echoes broader trends across the Indo-Gangetic Plain. [Bren d’Amour et al. \(2017\)](#) estimate that South Asia lost over 6 million ha of cropland to urban expansion between 2000 and 2030, with northern India being a primary hotspot. The concentration of urban growth in the Ludhiana–Amritsar–Jalandhar triangle suggests that industrial and commercial agglomeration economies continue to drive spatial expansion. Without land use regulation, continued encroachment threatens Punjab’s agricultural productivity, which already faces groundwater depletion from paddy cultivation ([Rodell et al., 2009](#)).

9.2 Forest and Vegetation Dynamics in Uttarakhand

The Vegetation → Cropland and Vegetation → Built-up transitions in Uttarakhand are concentrated in the Terai districts. This pattern is consistent with the long-standing process of forest-to-agriculture conversion in the Terai-Bhabar belt, driven by migration from the hills and government land settlement schemes ([Singh et al., 2017](#)). The relatively stable hill districts suggest that the Protected Area network (Jim Corbett NP, Rajaji NP, Kedarnath WLS) is effective in containing deforestation in the upper catchments, consistent with findings by [Joshi et al. \(2021\)](#).

9.3 Infrastructure-Driven Urbanization

The positive road–urban growth correlation in both states aligns with global evidence that road construction is a primary proximate cause of land cover change (Laurance et al., 2014). The policy implication is that new road projects in ecologically sensitive Terai districts of Uttarakhand should be accompanied by rigorous Environmental Impact Assessments and green corridor planning to prevent fragmentation.

9.4 Methodological Considerations

Several limitations should be acknowledged:

- **Dynamic World accuracy:** The Dynamic World model has reported overall accuracies of $\sim 75\%$ at 10 m resolution (Brown et al., 2022). Misclassification of fallow cropland as Barren or Vegetation is a known issue, particularly in semi-arid contexts.
- **Modal temporal aggregation:** Using the annual mode suppresses intra-annual variability but may introduce errors in areas with multiple cropping systems where no single class dominates across the year.
- **OSM completeness:** OSM road data in rural areas may be less complete than in urban centres, potentially underestimating road density in peripheral districts.

10 Conclusion

This study presents a comprehensive multi-temporal LULC analysis of Punjab and Uttarakhand for the period 2016–2025, using Google Earth Engine, Google Dynamic World V1, OSM road network data, and Python-based post-processing. The key findings are:

1. **Punjab** is undergoing sustained agricultural-to-urban land conversion, with Crop-land $\rightarrow$ Built-up as the dominant transition. Ludhiana leads in both absolute change and intensity index score.
2. **Uttarakhand** is experiencing forest and cropland conversion in its Terai belt, while hill districts remain stable. Haridwar and Udham Singh Nagar are the primary change hotspots.
3. Change in most high-ranked districts is **gradual** (consistent across sub-periods), but several districts in Uttarakhand show **ABRUPT** character, suggesting discrete events or policy-driven changes post-2020.
4. **Road density** shows a positive correlation with urban expansion in both states, consistent with global literature.

5. **Noise validation** via both quantitative thresholds and GEE seasonal cross-checks confirms that flagged changes in major districts are genuine and not classification artefacts.

The methodology developed here—combining GEE cloud computing, annual modal compositing, transition encoding, and Python statistical analysis—provides a replicable, scalable framework for district-level LULC monitoring across India. Future work could extend the analysis to include carbon stock changes, biodiversity impact assessments, and predictive modelling of land cover under different development scenarios.

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